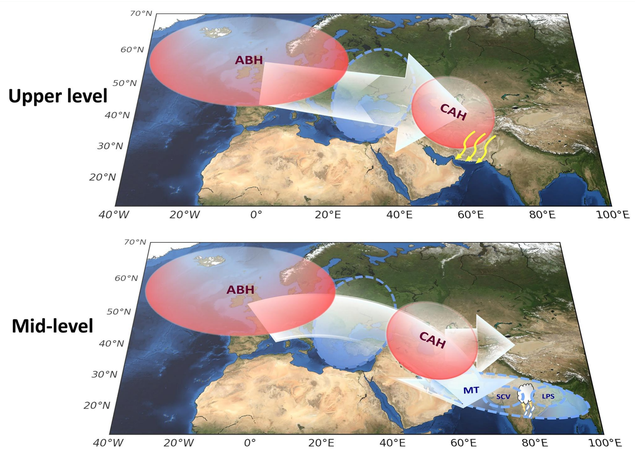
Revisiting 2025 Himalayan Extremes Through Tropical-Extratropical Interactions

[1]Disha Virmalwar; [1]Akshaya Nikumbh; [1]Anirudh Peri
[1]Center for Climate Studies, IIT Bombay,
*Correspondence to 30006814@iitb.ac.in

1. Introduction and Motivation

Extreme rainfall events in the **Himalayan region** during **2025** present a compelling case study for understanding tropical-extratropical interactions during the Indian summer monsoon. Past literature demonstrates that **Rossby wave breaking (RWB)**, characterized by equatorward intrusion of high potential vorticity (PV ≥1.5 PVU) air masses, can trigger organized convection and extreme precipitation. However, the dynamical mechanisms enabling such intrusions remain unclear. Previous work by Nikumbh et. al. (2023) has established that quasi-stationary atmospheric blocking highs over **Central Asia** facilitate RWB events by creating persistent flow patterns that allow extratropical PV streamers to penetrate equatorward. Building on this framework, this study examines four documented extreme rainfall events in 2025, investigating the co-occurrence of RWB signals and blocking highs days prior to local extremes, thereby establishing atmospheric blocking as a key mechanism linking tropical and

ABH: Atmospheric Blocking High | **CAH** : Central Asian High | **RWB** : Rossby Wave Breaking | **MT**: Monsoon Trough

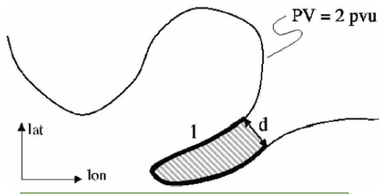


Nikumbh et. al. (2023)

2.Data and Methodology

2.1 Rossby Wave Breaking

- Potential Vorticity at 330K, 350K
- 1980-2019,2025 ERA5
- Dynamical Tropopause = **1.5 PVU**
- l : Contour length (**>1500 km**)
  d = geometric distance (**< 800 km**)



Streamer and Cut off Index
(Wernli and Sprenger 2007)

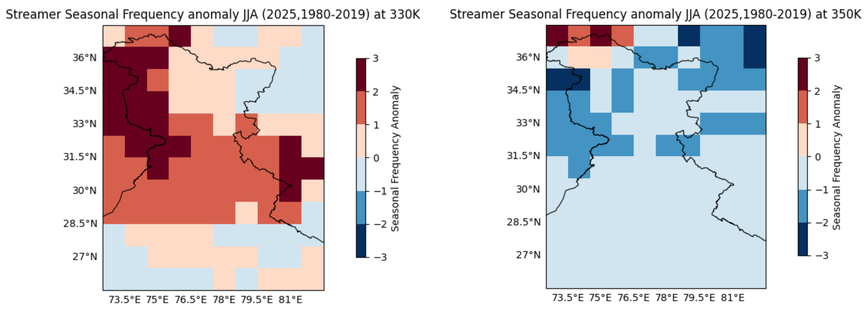
2.2 Atmospheric Blocking High

- Geopotential height anomalies at 200 hPa
- 1980-2025 ERA5
- Contour Tracking (ConTrack Algorithm)

Blocking High	ABH	CAH
Threshold	95th percentile	30
Persistence	2 days	2 days
Region	10°–80°N, 70°W–10°E	30°–60°N, 40°–90°E

3. Preliminary Results

3.1 Streamer Frequency Anomaly percentage of **2025** relative to **1980-2019 (JJA)**



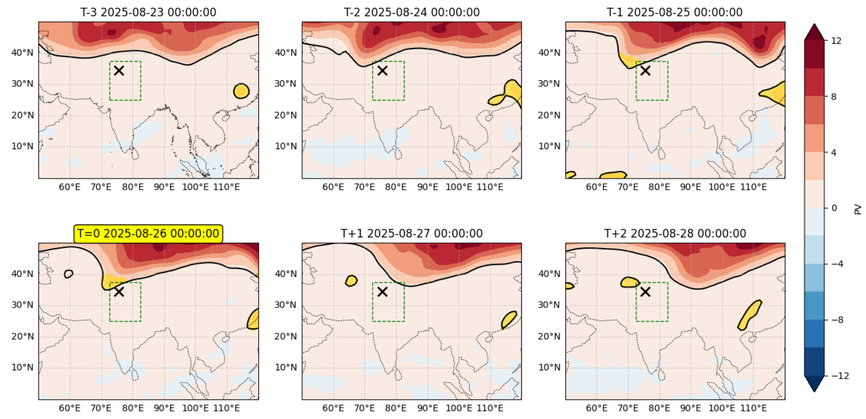
Relative enhancement of RWB streamers during JJA 2025 is slightly pronounced at 330K compared to climatological base line (1980-2019)

3.2 2025 Monsoon High-Impact Extreme Precipitation Events Over Himalayan Region

Date	Disaster Subtype	Origin	Location
2025-06-20	Flash flood	Monsoonal rains	Himachal Pradesh, Uttarakhand
2025-08-05	Mudslide	Violent cloudburst	Dharali village (Uttarkashi district)
2025-08-14	Flash flood	Heavy rain	Kishtwar (Jammu and Kashmir state)
2025-08-26	Landslide (wet)	Heavy rain	Jammu and Kashmir region
2025-08-15	Flood (General)	Monsoon rain	Punjab state

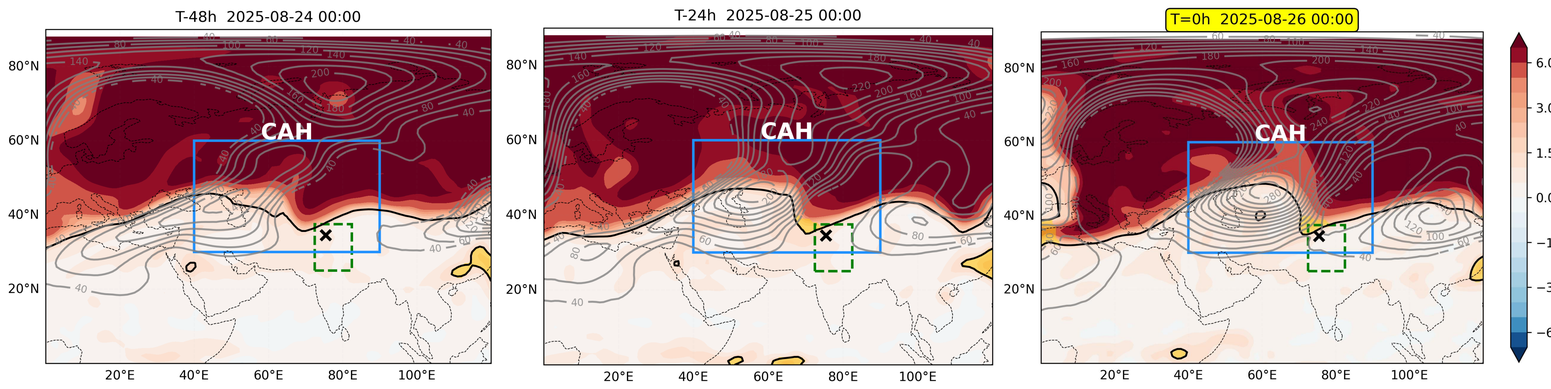
Data Source : https://www.emdat.be/

3.3 Jammu & Kashmir Landslide (26-08-2025)



Equatorward stratospheric PV streamer intrusions at **350 K** were detected **west** of the high-impact locations on the event day and preceding day

3.4 RWB at 350K and CAH at 200hpa during J&K Landslide



Presence of the **Central Asian High (CAH)** promoted recurving of the Rossby wave pathway, favoring Rossby Wave Breaking **(RWB)** prior to the extreme event

Summary and Discussion

1. JJA 2025 exhibited anomalously enhanced stratospheric streamer activity relative to the 1980-2019 climatology, particularly at the 330 K isentropic level
2. One of the high-impact event strongly aligned with the proposed framework, with convection developing along the eastern flank of the equatorward streamer.
3. Temporal evolution of the Central Asian High coincided with Rossby wave recurving and RWB development prior to the event.
4. Preliminary results suggest that tropical–extratropical interactions may play an important role in modulating Himalayan extreme precipitation events.
5. Future Work : a. Perform statistical and quantitative analyses to investigate the dynamical linkage between RWB, CAH variability, ENSO conditions and High impact extreme events

References

1. Nikumbh, A. C., Thakur, A. B. S., Chakraborty, A., Bhat, G. S., & Sukhatme, J. (2023). The role of the North Atlantic blocking high during large-scale heavy rainfall events over Central India. *Journal of the Atmospheric Sciences*, *80*(7), 1815-1827.2.
2. Wernli, H., & Sprenger, M. (2007). Identification and ERA-15 climatology of potential vorticity streamers and cutoffs near the extratropical tropopause. *Journal of the atmospheric sciences*, *64*(5), 1569-1586.
3. *Steinfeld, D., 2020: ConTrack - Contour Tracking. GitHub, https://github.com/steidani/ConTrack*.
4. Kaderli, S., 2023. WaveBreaking - Detection, Classification and Tracking of Rossby Wave Breaking. https://doi.org/10.5281/zenodo.14214463